Calcurion Maths Worksheets

GCSE Mathematics

Topic: Quadratic Equations

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Solve $x^2 - 5x + 6 = 0$. **[3]**

2. Solve $x^2 + 7x + 12 = 0$. **[3]**

3. Solve $x^2 - 9 = 0$. **[2]**

4. Solve $x^2 - 4x = 0$. **[2]**

5. Solve $2x^2 - 5x - 3 = 0$. **[4]**

6. Solve $3x^2 + x - 2 = 0$. **[4]**

7. Factorise $x^2 + 8x + 15$. **[2]**

8. Factorise $x^2 - 9x + 20$. **[2]**

9. Factorise $x^2 - 16$. **[2]**

10. Factorise $2x^2 + 7x + 3$. **[3]**

11. Solve $x^2 - 4x - 1 = 0$ giving exact answers. **[3]**

12. Solve $2x^2 + 3x - 1 = 0$ giving exact answers. **[4]**

13. Complete the square: $x^2 + 6x + 5$. **[3]**

14. Complete the square: $x^2 - 8x + 3$. **[3]**

15. Find the discriminant of $x^2 - 5x + 6 = 0$ and state the nature of the roots. **[3]**

Section B – Hard (Q16–25)

16. Solve $x^2 - 6x + 4 = 0$ by completing the square. **[4]**

17. Find the minimum value of $y = x^2 - 4x + 7$. **[3]**

18. Find the maximum value of $y = 5 - 2x - x^2$. **[4]**

19. Sketch $y = (x - 2)(x + 3)$, showing intercepts. **[4]**

20. Sketch $y = x^2 - 2x - 8$, showing intercepts and turning point. **[5]**

21. The equation $x^2 - 2x + k = 0$ has equal roots. Find k. **[3]**

22. Find the range of values of k for which $x^2 + kx + 4 = 0$ has real roots. **[4]**

23. Solve the simultaneous equations $y = x^2 - 3x + 2$ and $y = 2x - 1$. **[5]**

24. Solve $x^4 - 5x^2 + 4 = 0$. **[5]**

25. Prove that $x^2 + 4x + 5 > 0$ for all real x. **[4]**

End of Worksheet • *Show all working clearly.*